Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/17/2004
PI Name:	Schiffer, Peter Ph.D.		
Project Title:	Correlation Effects in Dense Wet and Dry Granular Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Agui, J. H.	**Contact Phone:**	216-433-5409
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA33G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This grant is supporting research into the fundamental properties of granular media. There is a particular focus on the properties of dense granular media and the development of correlations therein. A better understanding of the development and evolution of correlations in dense granular media. Granular media are used in a wide range of industrial processes, and understanding their properties is important to improving these processes. NASA missions (such as setting up lunar or martian bases) will also require extensive handling of granular materials, and therefore the understanding of these materials will be important for the design and implementation of those missions. Finally, the understanding of granular materials is of great fundamental scientific importance.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	More efficient industrial processes and better understanding of fundamental science always improves life on earth. This research should yield advances in both.
Task Progress:	In the first year of this research program we investigated the process of penetration of a solid object through a granular material. This process is relevant to a range of applications and is also of fundamental interest, since the required penetration force is an excellent example of the phenomenon of "local jamming." We measured the penetration force required to push a flat plate vertically downward through a dense granular medium, focusing in particular on the effects of the bottom boundary of the vessel containing the medium. Our data demonstrate that the penetration force near the bottom is strongly affected by the surface properties of the bottom boundary, even many grain diameters above the bottom. Furthermore, the data indicate an intrinsic length scale for the interaction of the penetrating object with the vessel bottom via the medium. This length scale, which corresponds to the local jamming induced by penetration, has a square root dependence both upon the size of the penetrating object and the ambient granular stress in the vicinity of the bottom of the vessel containing the granular medium, but it is independent of penetration velocity and grain diameter. In addition to the above investigation, we have also begun three other experimental projects to probe the nature of correlations in granular materials. One is a study of the penetration force of a solid object pushed vertically upward from the bottom of a granular medium. This will probe the effects of a free surface on the penetration force and local jamming. A second study is in investigation of the effects of thermal annealing on granular packing. A third experiment is on the flow of grains through an orifice in a shaken granular sample.
Bibliography Type:	Description: (Last Updated: 11/15/2007)
Articles in Peer-reviewed Journals	Stone MB, Bernstein DP, Barry R, Pelc MD, Tsui YK, Schiffer P. "Stress propagation: getting to the bottom of a granular medium." Nature. 2004 Feb 5;427(6974):503-4. PMID: 14765185 , Feb-2004
Articles in Peer-reviewed Journals	Stone MB, Barry R, Bernstein DP, Pelc MD, Tsui YK, Schiffer P. "Local jamming via penetration of a granular medium." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Oct;70(4 Pt 1):041301. Epub 2004 Oct 29. PMID: 15600401 , Oct-2004
Presentation	Schiffer, P. "Drag, Penetration and Local Jamming in Granular Media" Gordon Conference on Granular and Granular Fluid Flows (invited short talk), Colby College, ME, July 1, 2004. Jun-2004
Presentation	Schiffer, P. "Local Jamming: Drag and Penetration Through Granular Media" Center for Nonlinear Systems Seminar, Physics Dept., University of Texas, Austin, TX, November 10, 2004. Nov-2004
Presentation	Schiffer, P. "Local Jamming: Drag and Penetration Through Granular Media" Condensed Matter Physics Seminar, Physics Dept., Johns Hopkins University, Baltimore, MD, December 1, 2004. Dec-2004